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SPHERE CARE

Video and AI Aged Care Platform

Project Marketing

CSIT321 Software Engineering Capstone | University of Wollongong
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1. Introduction

This document presents the marketing case for Sphere Care, an AI-assisted care communication and documentation platform designed for aged care facilities in Australia. It covers the problem being solved, the target market and user personas, the proposed solution, the design approach and the commercial roadmap.

The purpose of this section is to show how Sphere Care and create value for aged care facilities by reducing documentation workload, improving communication with families, supporting safer care review and helping facilities maintain clearer records for accountability and compliance.
The intended audience includes the project supervisor, subject coordinator, potential aged care facility partners and future stakeholders who may evaluate, adopt or extend the platform.

2. Elevator Pitch

> Sphere Care is an AI-assisted care documentation and communication platform built for aged care facilities. It helps care staff record care interactions, generate searchable transcripts and summaries, manage resident updates, communicate with families and review potential care issues through one secure, role-based system.

3. The Problem

Australia's aged care sector is facing increasing pressure as the population ages, care needs become more complex, and regulatory requirements continue to expand. Facilities must deliver safe, person-centred care while managing workforce shortages, growing documentation demands, and higher expectations from residents and families. Despite advances in healthcare technology, many aged care providers still rely on fragmented systems and manual processes that reduce efficiency and create operational challenges.

Sphere Care was developed to address these challenges by providing an integrated platform that improves documentation, communication, safety monitoring, and compliance management.

3.1 Industry Challenges

The Australian aged care industry is undergoing significant transformation. Demand for aged care services continues to grow as the proportion of older Australians increases, placing additional pressure on care providers and staff. At the same time, facilities must comply with stricter quality standards, maintain comprehensive records, and demonstrate transparency in their care practices.

These challenges create a difficult balance between delivering high-quality resident care and managing administrative responsibilities. Many facilities operate with limited staffing resources, particularly during evenings and overnight shifts, making it increasingly important to adopt technology that improves efficiency without disrupting existing workflows.

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3.2 Documentation Burden
Care staff spend a significant amount of time documenting resident interactions, medication administration, wellbeing observations, incident reports, and daily care activities. In many facilities, this information is entered manually into multiple systems or recorded after the event has occurred.

Manual documentation creates several problems:

- Increased administrative workload for care staff.
- Reduced time available for direct resident care.
- Inconsistent record quality between staff members.
- Delayed recording of important care events.
- Difficulty locating historical information when needed.

As a result, valuable staff time is often spent completing paperwork rather than supporting residents. Facilities need more efficient methods for capturing and organising care information while maintaining record accuracy.

3.3 Safety Monitoring Gaps
Resident safety is a critical responsibility for aged care providers. However, care staff cannot continuously observe every resident at all times, particularly during busy periods, overnight shifts, or in facilities with limited staffing levels.

Important events such as falls, unusual movement patterns, signs of distress, agitation, wandering behaviour, or medication refusal may not always be identified immediately. Delays in recognising these situations can affect response times and increase risk for vulnerable residents.

Many existing systems depend primarily on manual observation and incident reporting, providing limited support for proactive monitoring. This creates an opportunity for AI-assisted safety technologies that can help staff identify potential risks earlier and prioritise appropriate responses.

Sphere Care addresses this challenge through SCVAM (Sphere Care Video Analysis Module), which assists staff by analysing recorded video data and generating reviewable safety flags for potential care-related events.

3.4 Communication and Accessibility Gaps
Effective communication is essential in aged care environments, yet many residents face barriers that make communication difficult. Some residents may experience hearing impairments, speech limitations, cognitive decline, or other conditions that affect their ability to express needs clearly.

Traditional care systems are largely designed around written notes and verbal communication. As a result, residents who rely on gestures, visual communication, or sign-based interaction may face additional challenges when communicating with care staff.

Communication difficulties can contribute to misunderstandings, reduced independence, delayed responses to resident needs, and lower overall quality of care.

Sphere Care addresses this challenge through ASLLM (AI Sign Language and Gesture Communication Module), which assists in recognising selected sign-based and gesture-based communication inputs and converting them into readable outputs that can be reviewed by staff. This helps support more inclusive communication and improves accessibility for residents who may struggle with traditional communication methods.

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3.5 Compliance Pressures

Following recommendations from the Royal Commission into Aged Care Quality and Safety, aged care providers face increasing expectations regarding accountability, transparency, and documentation quality. Facilities must demonstrate that care activities have been properly performed, recorded, reviewed, and stored.

Key compliance challenges include:

- Maintaining complete and accurate care records.
- Producing evidence during audits and reviews.
- Managing consent and privacy requirements.
- Tracking user actions and system changes.
- Ensuring secure storage of sensitive information.

Preparing for audits can be time-consuming when information is distributed across multiple systems. Facilities require reliable audit trails, searchable records, and structured reporting capabilities to support compliance activities efficiently.

Australia's aged care sector faces a growing accountability crisis. The Royal Commission into Aged Care Quality and Safety (Royal Commission into Aged Care Quality and Safety 2021) found systemic failures in care documentation, family communication, and incident detection. Despite these findings, most facilities still rely on:

- Manual care notes that are time-consuming to write and difficult to search.
- Fragmented communication between staff, families, and administrators.
- Delayed family updates that can reduce transparency and trust.
- Care records distributed across multiple systems and documents.
- Limited visibility of potential safety-related events during busy or overnight shifts.
- Increasing compliance obligations requiring accurate, auditable, and secure documentation.

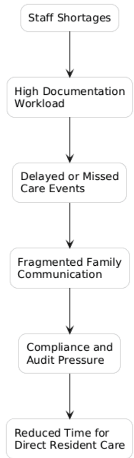

Figure 3.5 - Problem Landscape

As a result, care staff often spend valuable time completing administrative tasks instead of supporting residents directly. Family members may struggle to stay informed about the wellbeing of loved ones, while facility managers face growing pressure to demonstrate accountability and compliance.

4. Target Market Analysis

Sphere Care is designed for the Australian aged care sector, a market experiencing significant growth due to population ageing, increasing care complexity, and expanding regulatory requirements. Residential aged care providers are under pressure to improve care quality, strengthen documentation practices, enhance communication with families, and maintain compliance with evolving industry standards.

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As facilities face ongoing workforce shortages and administrative challenges, there is increasing demand for technology solutions that improve operational efficiency without increasing staff workload. Sphere Care addresses this need by providing a practical, web-based platform that combines documentation, communication, safety monitoring, and compliance support within a single ecosystem.

4.1 Australian Aged Care Market

Australia's aged care industry supports more than 240,000 residents across approximately 2,700 residential aged care facilities nationwide. The sector is expected to continue expanding as Australia's ageing population grows and demand for long-term care services increases.

- At the same time, aged care providers face several industry-wide challenges:
- Growing resident populations with increasingly complex care needs.
- Workforce shortages across clinical and support roles.
- Increasing documentation and compliance obligations.
- Greater expectations for transparency and accountability.
- Demand for improved communication between facilities and families.
- Rising pressure to improve operational efficiency while maintaining care quality.

These challenges create a strong opportunity for digital platforms that can help providers reduce administrative burden, improve visibility of care activities, and strengthen compliance processes.

6.2 Market Segments

Sphere Care is designed to support multiple aged care market segments, each with distinct operational requirements and challenges.

Segment	Description	Pain Points Addressed
Private residential aged care facilities	The largest market segment; profit-driven and strongly compliance-focused	Audit-ready documentation, AI-supported safety monitoring, and improved family communication
Not-for-profit aged care operators	Community-focused organisations with limited budgets	Affordable web-based deployment and no per-seat licensing
Government-operated facilities	Facilities with strict regulatory obligations and complex procurement processes	Immutable audit logs, compliance reporting, and per-facility data isolation
Aged care at-home services	A growing service model with fragmented documentation and communication	Mobile-first recording, centralised records, and family communication support

By supporting multiple deployment scenarios, Sphere Care can address both current and future opportunities within the broader aged care market.

4.3 Go-to-Market Focus

The initial go-to-market strategy focuses on small-to-medium private residential aged care facilities located in New South Wales and Victoria. These facilities typically support between 50 and 200 residents and often experience the greatest pressure to improve documentation quality, family communication, and compliance management while operating with limited resources.
This segment was selected because:

- It represents a large and accessible market.
- Facilities face significant accreditation and regulatory pressure.
- Many providers cannot justify the cost of large enterprise healthcare systems.
- Adoption barriers are lower for web-based solutions requiring minimal infrastructure.
- Demonstrating value within this segment creates opportunities for broader expansion.

Following successful adoption within residential aged care facilities, Sphere Care can expand into not-for-profit organisations, government-operated facilities, disability support services, and home-care providers.

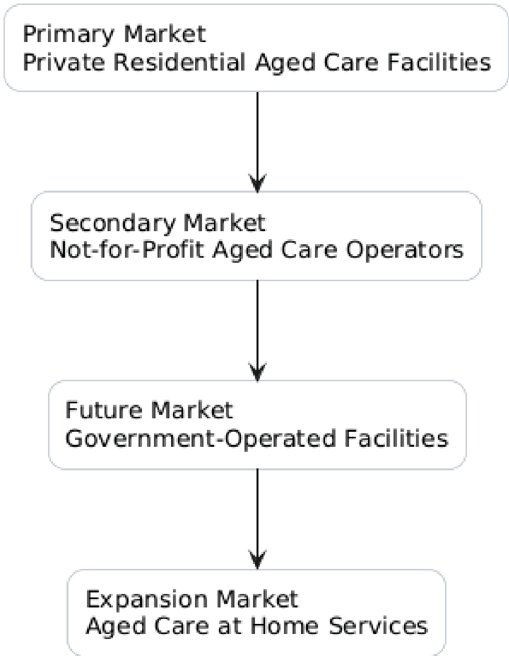

Figure 4.1 - Target Customer Groups

The Sphere Care customer ecosystem consists of four primary stakeholder groups. Residential aged care providers represent the primary purchasing decision-makers, while care staff, family members, and compliance stakeholders influence adoption and ongoing platform value.

By addressing the needs of each stakeholder group simultaneously, Sphere Care creates a platform that delivers operational, clinical, communication, and compliance benefits across the entire aged care environment.

5. User Persona

The following personas represent the primary stakeholder groups that Sphere Care is designed to support. While fictional, they are based on common roles, responsibilities, and challenges found within the Australian aged care sector. These personas help illustrate how different users interact with the platform and the value that Sphere Care provides across clinical, operational, communication, and compliance workflows.

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5.1 Sandra Thompson, 52 Director of Nursing

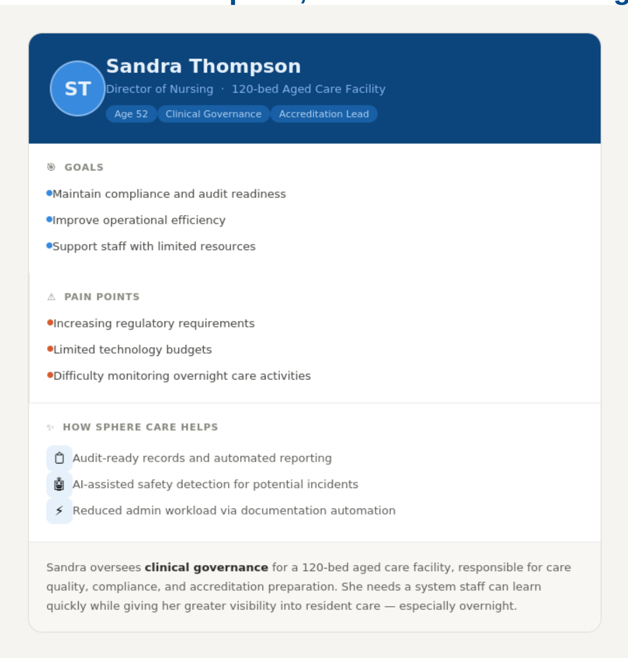

figure 5.1 Sandra Thompon's persona

5.2 James Wilson, 34 Registerd Nuese

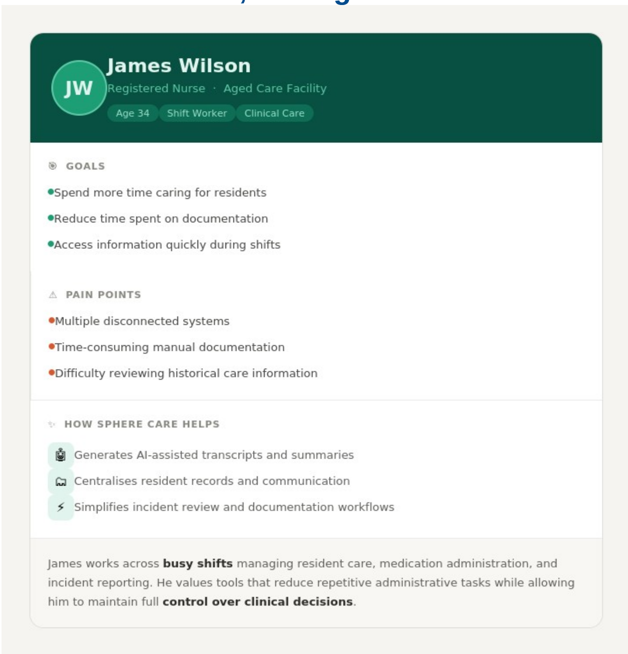

figure 5.2 James Wilson's persona

5.3 Linda Roberts, 61, Family Member

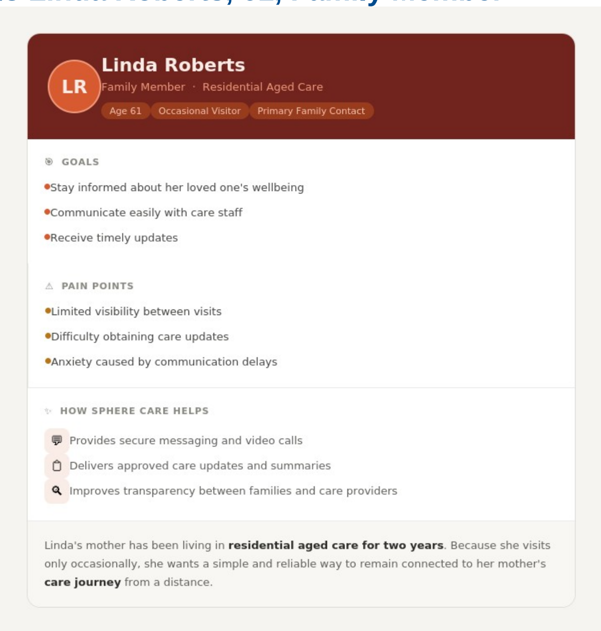

figure 5.3 Linda Robert's persona

5.4 Robert Thompson, 48, Compliance Officer

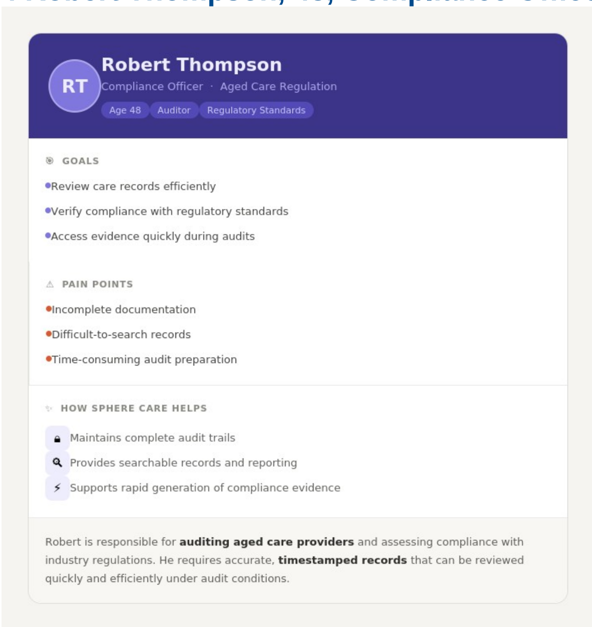

figure 5.4 Robert Thompson's Persona

The platform is designed to support a diverse range of users, including care staff, facility administrators, family members, and compliance stakeholders. Although each group has different objectives, all share a common need for accurate information, improved communication, reduced administrative burden, and greater confidence in care delivery.

Common Needs Across All Personas

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Although each stakeholder interacts with Sphere Care differently, they share several common requirements:

- Faster access to accurate information.
- Reduced administrative burden.
- Improved communication and transparency.
- Enhanced resident safety.
- Stronger compliance and accountability.
- Greater confidence in care delivery processes.

Sphere Care addresses these needs through a unified platform that combines AI-assisted documentation, safety monitoring, communication tools, resident management, and audit-ready reporting within a secure and accessible ecosystem.

6. The Solution: Sphere Care

Sphere Care transforms fragmented aged care workflows into a connected, intelligent, and user-centred care ecosystem. Rather than relying on multiple disconnected systems for documentation, communication, incident management, and compliance reporting, Sphere Care brings these functions together within a single integrated platform.

The solution combines AI-assisted documentation, safety monitoring, communication services, resident management, and compliance support to help aged care providers operate more efficiently while maintaining high standards of care. By reducing administrative burden and improving information accessibility, Sphere Care enables staff to focus more time on residents and less time on paperwork.

A key differentiator of the platform is its ability to combine advanced technologies with human oversight. Artificial intelligence is used to assist staff rather than replace them, ensuring that professional judgement remains central to all care decisions.

6.1 Platform Overview

Sphere Care consists of two primary user-facing applications supported by a secure backend infrastructure and AI intelligence layer.

The **Staff Web Platform** provides care workers, nurses, administrators, and facility managers with access to resident records, AI-assisted documentation tools, safety alerts, communication functions, reporting capabilities, and operational dashboards.

The **Family Mobile Application** allows authorised family members to communicate with care providers, participate in secure video calls, receive updates, and remain connected to their loved one's care journey.

Supporting both applications is the Sphere Care backend, which manages authentication, resident records, communication services, notifications, audit logging, and data storage. Integrated within this architecture are two specialised AI modules:

- **SCVAM (Sphere Care Video Analysis Module)**, which assists staff by analysing recorded video content and generating reviewable safety-related flags.
- **ASLLM (AI Sign Language and Gesture Communication Module)**, which assists residents who rely on gesture-based communication by converting recognised signs into readable outputs for staff review.

Together, these components create a unified platform that improves visibility, communication, documentation quality, and operational efficiency across aged care environments.

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6.2 Core Value Proposition

Sphere Care delivers value by helping aged care providers spend less time managing administration and more time supporting residents.

The platform addresses five major challenges identified throughout this report:

1. Documentation burden.
2. Safety monitoring limitations.
3. Communication barriers.
4. Accessibility challenges.
5. Compliance pressures.

To address these challenges, Sphere Care provides four key value outcomes:

Value Area	Benefit
Efficiency	Reduces manual documentation through AI-assisted transcription and summarisation.
Safety	Supports earlier identification of potential care-related events through SCVAM-assisted monitoring.
Accessibility	Improves communication opportunities for residents through ASLLM-supported gesture recognition.
Compliance	Provides searchable records, audit logs, and reporting tools that support accountability and regulatory requirements.

Unlike traditional systems that focus on a single aspect of care management, Sphere Care combines documentation, communication, resident engagement, safety monitoring, accessibility support, and compliance management within one integrated ecosystem.

This unified approach improves operational efficiency, strengthens transparency, enhances communication, and helps facilities provide safer, more responsive, and more inclusive care. Ultimately, Sphere Care enables organisations to focus on what matters most: delivering high-quality care while maintaining confidence in the accuracy, accessibility, and security of resident information.

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7. Product Ecosystem

Sphere Care is designed as a connected care ecosystem rather than a standalone software application. The platform integrates documentation, communication, resident management, safety monitoring, accessibility support, and compliance functions within a single environment. This approach eliminates the need for multiple disconnected systems and provides a more efficient workflow for staff, families, and facility administrators.

The ecosystem is built around the principle that care information should be accessible, secure, and connected throughout the entire resident journey. By bringing together communication tools, AI-assisted services, resident records, and governance controls, Sphere Care creates a unified platform that supports both operational efficiency and quality care delivery.

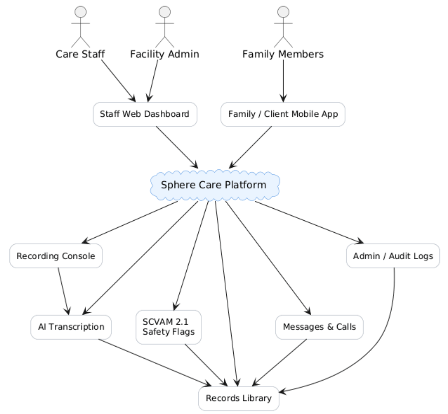


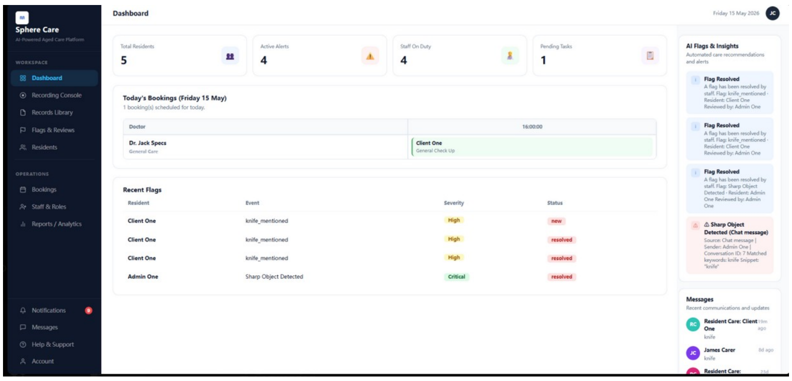
Figure 7 - Sphere Care Ecosystem

7.1 Staff Web Platform

The Staff Web Platform serves as the primary operational workspace for care workers, nurses, administrators, and facility managers. It centralises daily workflows and provides staff with immediate access to the tools required to manage resident care effectively.

Key functions include:

- Resident record management.
- AI-assisted recording and documentation.
- Safety flag review and escalation.
- Messaging and communication tools.
- Appointment and activity management.
- Analytics and operational reporting.
- Administrative and audit functions.



The platform is designed around a dashboard-based interface that reduces navigation complexity and supports fast access to critical information.

7.2 Mobile / Family Application

The Family Mobile Application improves communication and transparency between care providers and family members. It allows authorised users to remain informed about their loved one's care journey without requiring complex training or specialised software.

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Key functions include:

- Secure video and audio communication.
- Real-time messaging.
- Appointment booking and management.
- Notifications and updates.
- Access to approved resident information.
- Family engagement features.

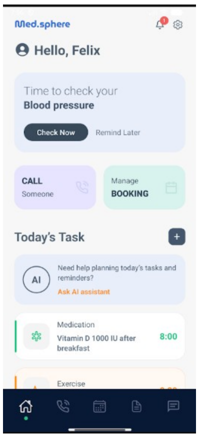

Figure 7.2 – Mobile Application Interface

The application is designed with accessibility and simplicity in mind, ensuring ease of use for users with varying levels of technical confidence.

7.3 AI Intelligence Layer

The AI Intelligence Layer provides the intelligent capabilities that differentiate Sphere Care from traditional aged care management systems. Rather than replacing staff judgement, the AI modules act as assistive tools that help staff document, review, and respond to care-related events more efficiently.
The layer consists of two primary components:

- **SCVAM (Sphere Care Video Analysis Module)**
  SCVAM assists staff by analysing recorded video content and identifying potential care-related events such as falls, unusual movement patterns, agitation, or medication refusal. Detected events are converted into reviewable safety flags that require human assessment before action is taken.
- **ASLLM (AI Sign Language and Gesture Communication Module)**
  ASLLM assists residents who rely on gesture-based communication by recognising selected signs and converting them into readable outputs. This helps improve accessibility and supports more inclusive communication between residents and care staff.

Together, these technologies reduce administrative workload, improve visibility of care activities, and support faster decision-making.

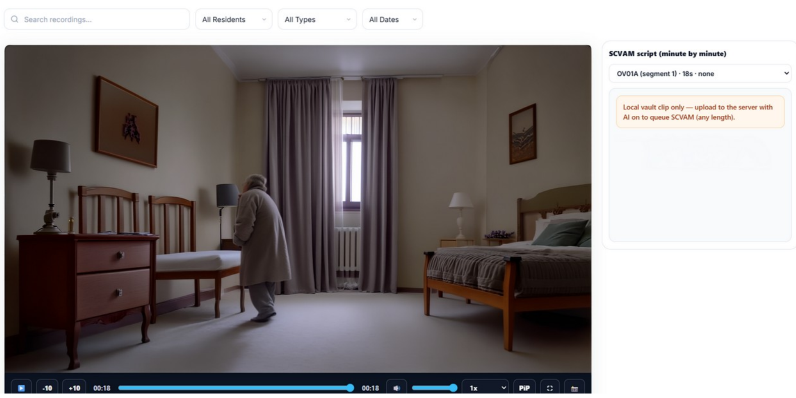

Figure 7.3a – AI Safety Monitoring

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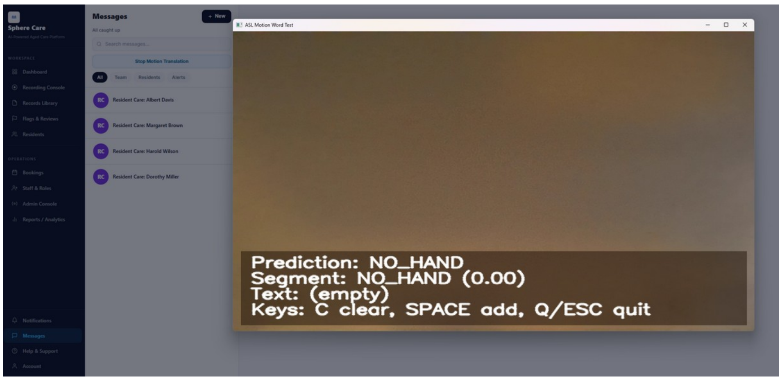


Figure 7.3b – Gesture Communication Support

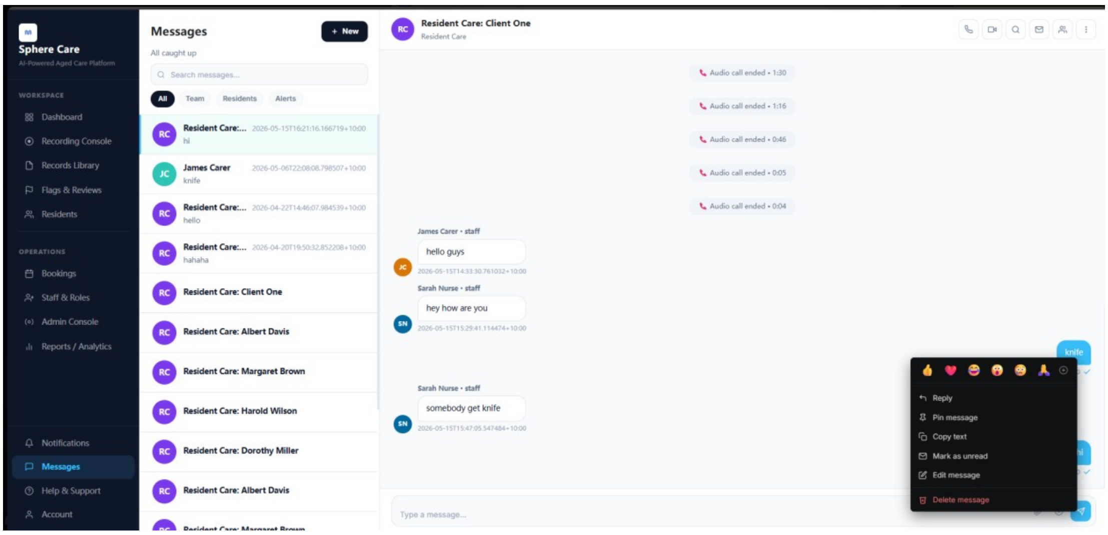
7.4 Communication Layer

Effective communication is critical in aged care environments. Sphere Care includes an integrated communication layer that connects staff, residents, and families through a secure and structured framework.

Communication services include:

- Real-time messaging.
- Video and audio calling.
- Resident-specific discussion channels.
- Notification management.
- Communication history tracking.

By consolidating communication into a single platform, Sphere Care reduces information fragmentation and improves collaboration across care teams.



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7.5 Compliance & Governance Layer

The Compliance and Governance Layer supports accountability, transparency, and regulatory compliance. Every significant action performed within the platform is recorded and managed through structured governance controls.

Key capabilities include:

- Audit logging.
- Role-based access control.
- Consent management.
- Secure data storage.
- Record retention management.
- Compliance reporting and export tools.

This layer helps facilities maintain audit readiness while protecting sensitive resident information and ensuring that access is restricted to authorised users.

8. Core Features

The core features of Sphere Care were developed directly from the challenges identified in the aged care sector. Each feature is designed to reduce administrative burden, improve communication, enhance resident safety, strengthen accessibility, and support regulatory compliance.

8.1 AI-Assisted Recording and Documentation

Staff can start a recording session with a single action. Sphere Care automatically generates transcripts and AI-assisted summaries, creating structured records that can be reviewed, searched, and stored securely.

Benefits

- Reduced documentation workload.
- Faster record creation.
- Improved record consistency.
- Searchable resident history.

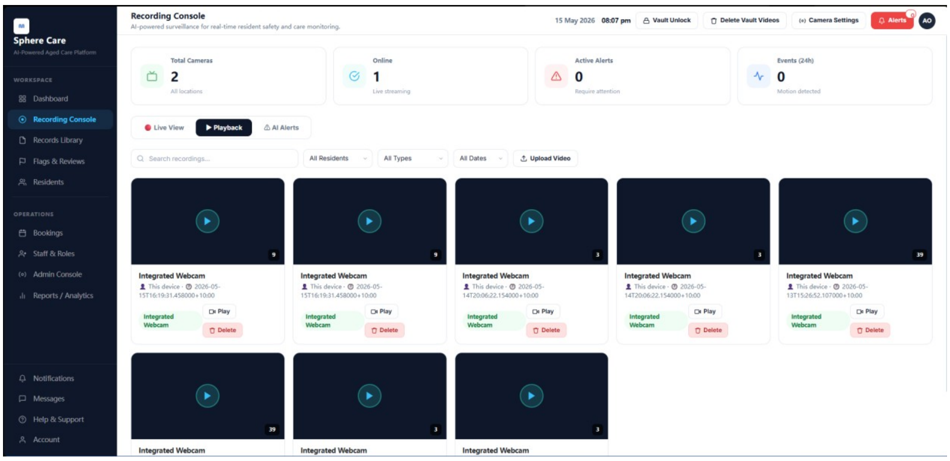

Figure 8.1 – Recording Console

8.2 AI-Assisted Safety Monitoring (SCVAM)

SCVAM analyses recorded care interactions and identifies potential care-related events requiring staff review. Rather than replacing staff judgement, the system provides additional visibility into events that may otherwise be overlooked.

Benefits

- Earlier identification of potential incidents.
- Improved overnight monitoring support.
- Faster review and escalation workflows.
- Enhanced resident safety.

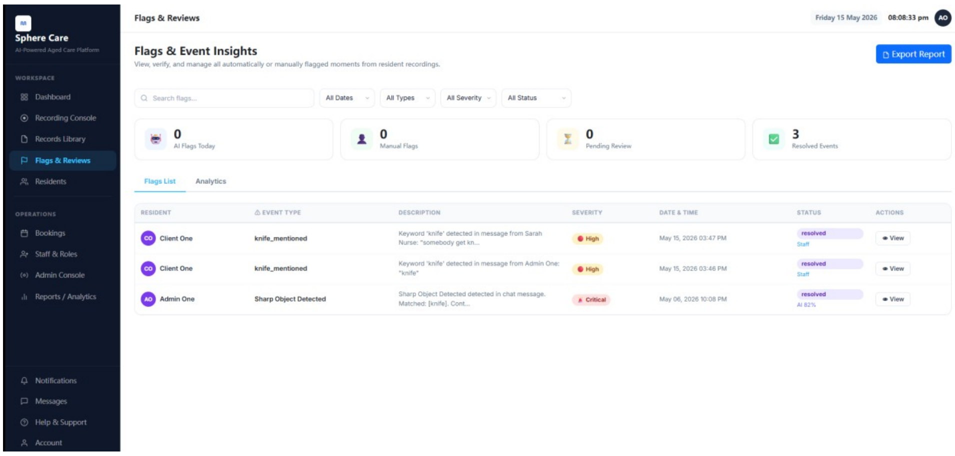

Figure 8.2 – Safety Flags and Reviews

8.3 Sign-Based Communication Support (ASLLM)

ASLLM assists residents who rely on gestures or sign-based communication by converting recognised signs into readable outputs that can be reviewed by staff.

Benefits

- Improved communication accessibility.
- Greater resident independence.
- More inclusive care delivery.
- Better understanding between residents and staff.

Figure 8.3 – ASLLM Communication Interface

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8.4 Secure Family Communication

Sphere Care enables secure messaging and video communication between families and care providers, improving transparency and engagement throughout the care journey.

Benefits

- Increased family involvement.
- Reduced communication barriers.
- Improved trust and transparency.

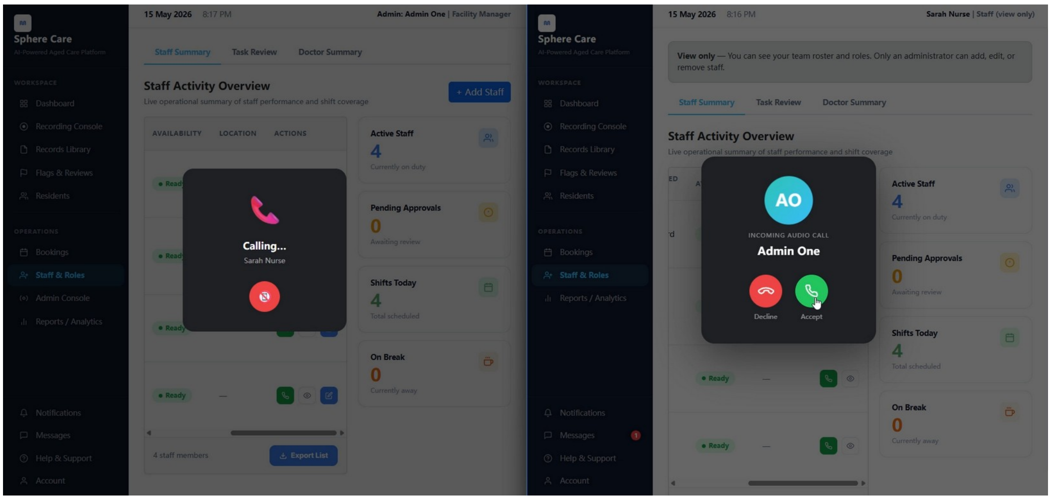

Figure 8.4 – Video Call Interface

8.5 Audit-Ready Compliance Management

Every significant system action is logged and traceable, helping facilities prepare for audits, reviews, and accreditation activities.

Benefits

- Improved accountability.
- Faster audit preparation.
- Better compliance reporting.

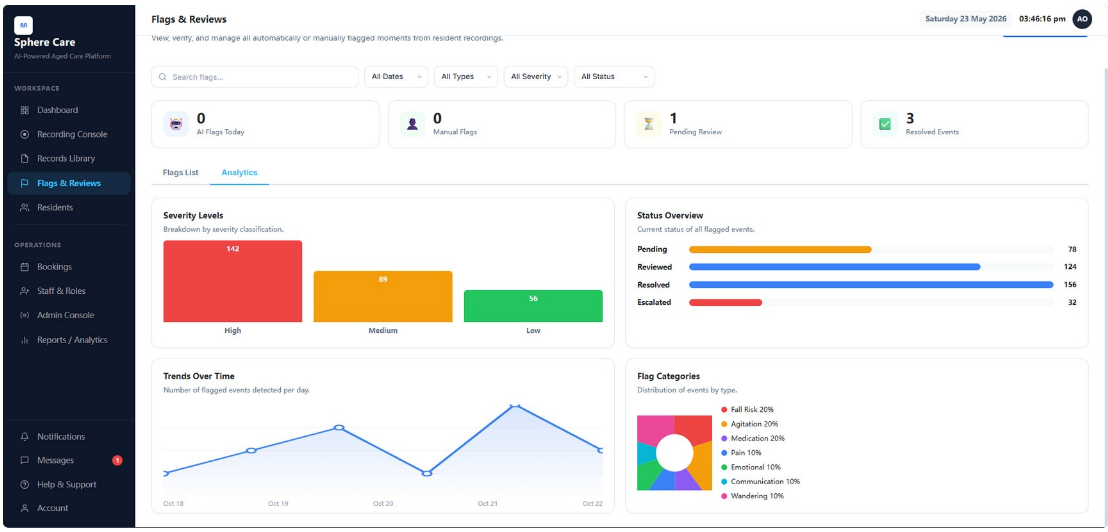

Figure 10.5 – Audit Log and Reporting

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9. Product Showcase

The following screenshots demonstrate how Sphere Care translates the needs identified throughout this report into a practical and user-friendly solution. Each interface was designed to support efficiency, accessibility, transparency, and compliance within aged care environments.

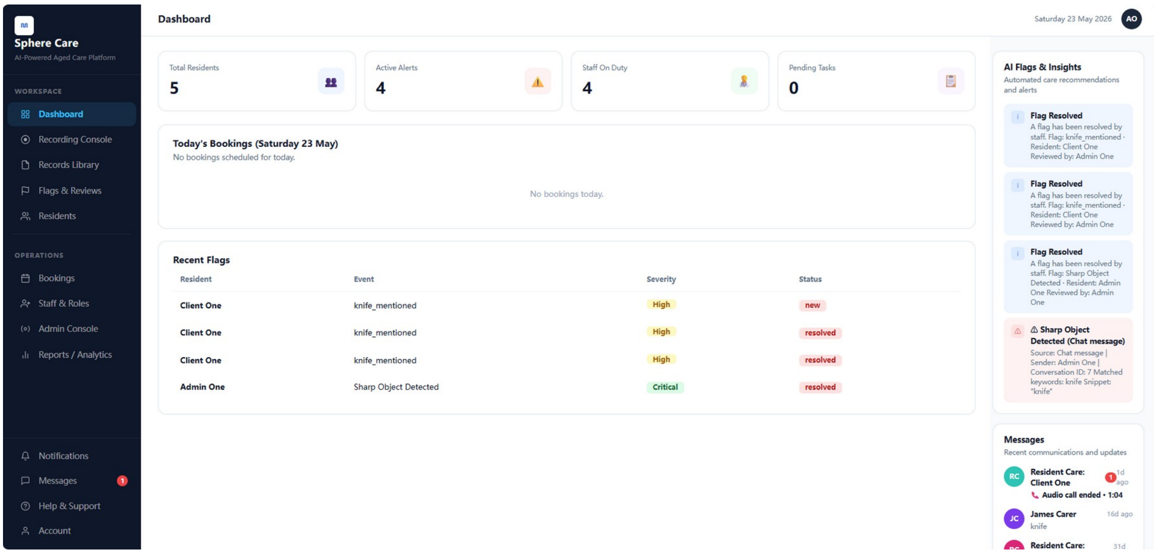
The product showcase highlights the key components of the platform and demonstrates how staff, families, and administrators interact with Sphere Care throughout the care journey.

9.1 Staff Dashboard

The Staff Dashboard provides a centralised overview of daily operations, resident activity, AI alerts, upcoming appointments, and key system metrics. It is designed to give staff immediate visibility of important information while reducing navigation complexity.

Key Benefits

- Centralised operational overview.
- Fast access to resident information.
- Visibility of important alerts and notifications.
- Improved workflow efficiency.



9.2 Recording Console

The Recording Console enables staff to capture care interactions and generate AI-assisted documentation. Recordings can be securely stored, transcribed, summarised, and linked directly to resident records.

Key Benefits

- Reduced manual note-taking.
- Faster documentation workflows.
- Improved record consistency.
- Searchable care history.

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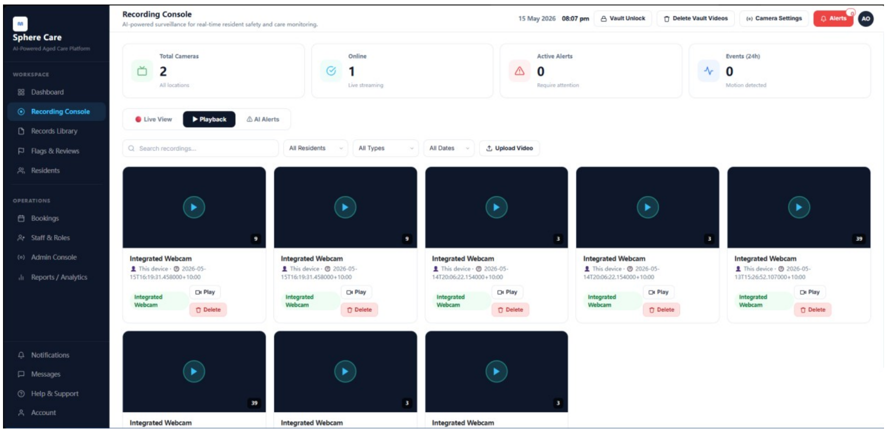

Figure 9.2 – Recording Console

9.3 Records Library

The Records Library provides secure access to resident recordings, transcripts, summaries, and historical care information. Staff can quickly locate records using filtering and search functions.

Key Benefits

- Centralised record management.
- Faster information retrieval.
- Improved continuity of care.
- Enhanced documentation visibility.

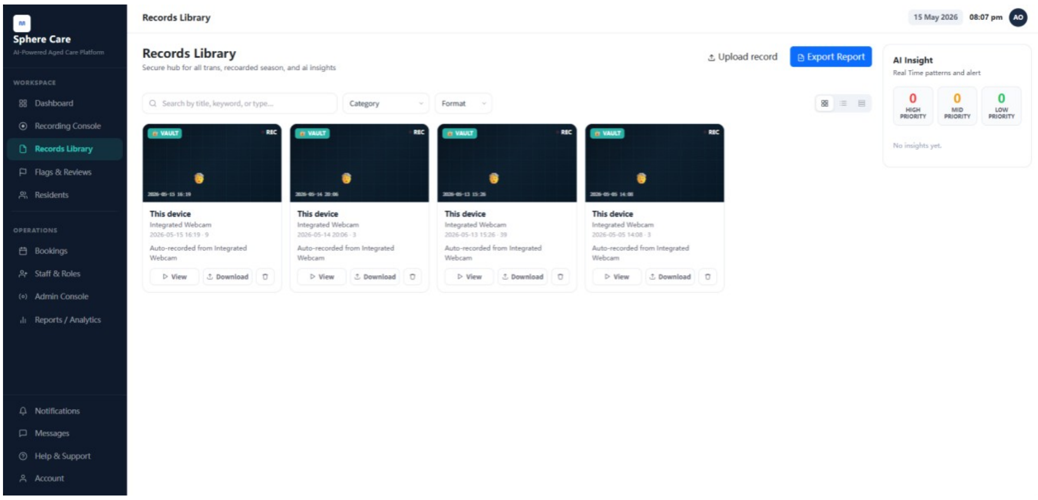

Figure 11.3 – Records Library

9.4 AI Flags and Reviews

The Flags and Reviews interface allows staff to review potential care and safety events identified by SCVAM. Events are prioritised according to severity and require human review before action is taken.

Key Benefits

- Earlier visibility of potential incidents.
- Improved review workflows.
- Supports safer care delivery.
- Maintains human oversight.

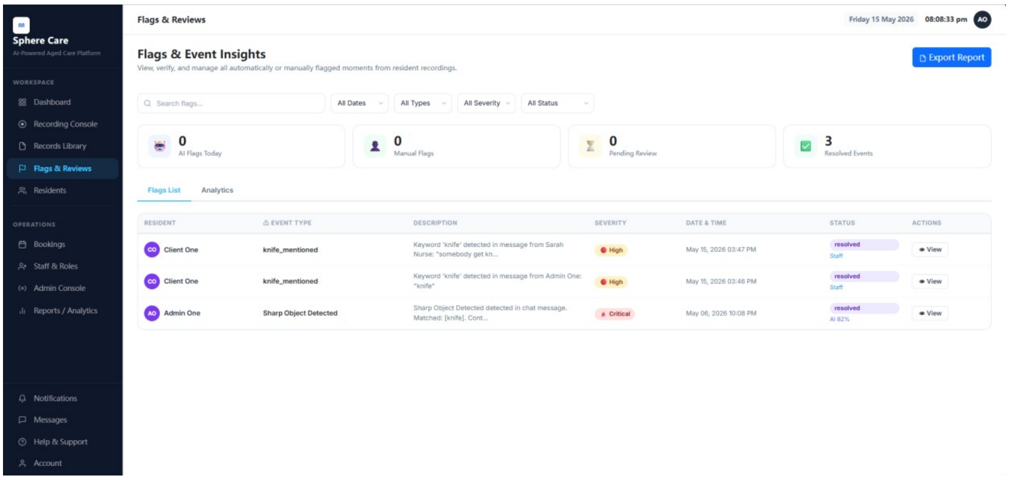

Figure 11.4 – AI Flags and Reviews

9.5 Resident Profile Management

The Resident Profile page provides a comprehensive view of resident information, care history, communication records, appointments, and associated documentation.

Key Benefits

- Complete resident overview.
- Centralised care information.
- Faster access to important records.
- Improved care coordination.

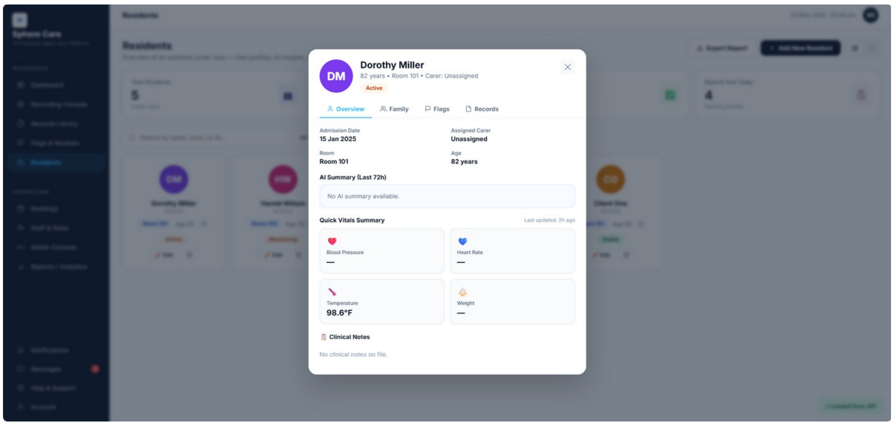

Figure 11.5 – Resident Profile

9.6 Mobile Video Calling

The Mobile Video Calling feature enables secure communication between residents, family members, and care providers. Calls can be initiated directly through the platform without requiring complex setup procedures.

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Key Benefits

- Improved family engagement.
- Increased communication accessibility.
- Reduced distance barriers.
- Greater transparency.

Figure 11.6 – Mobile Video Call

9.7 Messaging Centre

The Messaging Centre provides secure, resident-focused communication channels for staff and family members. Conversations are organised and linked to relevant resident records where appropriate.

Key Benefits

- Improved collaboration.
- Reduced information fragmentation.
- Faster communication.
- Centralised message history.

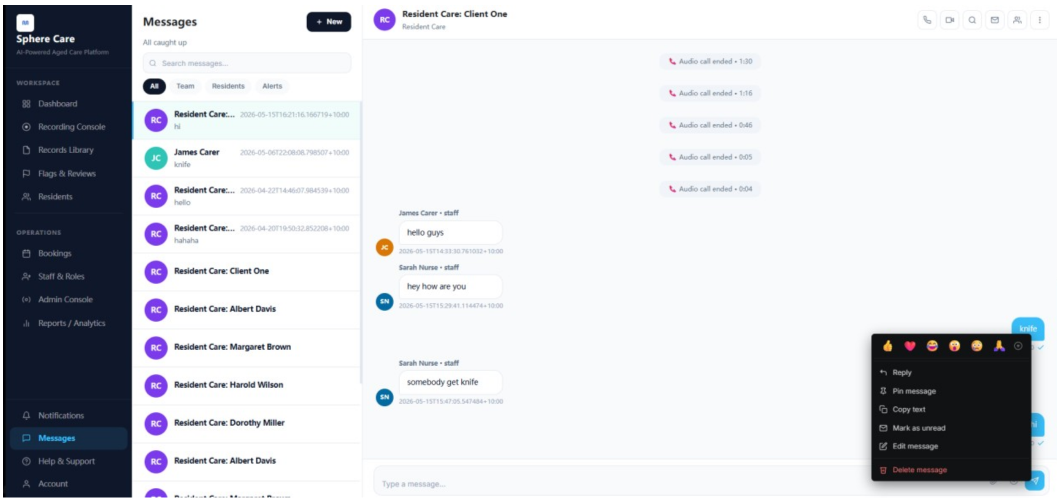

Figure 11.7 – Messaging Centre

9.8 Notifications and Alerts

The Notifications and Alerts interface keeps users informed about important events, appointments, messages, and AI-generated reviews requiring attention.

Key Benefits

- Faster awareness of important events.
- Improved response times.
- Better operational visibility.
- Enhanced user engagement.

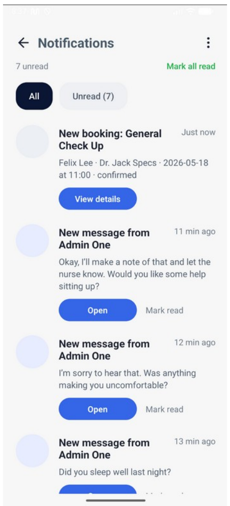

Figure 11.8 – Notifications and Alerts

Together, these interfaces demonstrate how Sphere Care combines AI-assisted documentation, safety monitoring, communication, accessibility support, and compliance management into a single integrated platform. By bringing these functions together within one ecosystem, Sphere Care helps aged care providers improve efficiency, strengthen transparency, and deliver higher-quality care while maintaining confidence in documentation and compliance processes.

10.Key Differentiators

Sphere Care was designed to address challenges that are often overlooked by traditional aged care management systems. Rather than focusing solely on record management or communication, the platform combines documentation, communication, accessibility support, safety monitoring, and compliance management within a single ecosystem.

Feature	Traditional Systems	Sphere Care
Care documentation	Manual text entry; time-consuming and inconsistent across staff	AI-assisted transcription and summarisation to reduce manual documentation effort
Safety Monitoring	Mainly depends on staff observation; limited visibility during busy or overnight periods	AI-assisted safety flagging for potential events such as falls, agitation or medication refusal
Safety Monitoring	Limited support for gesture-based communication	ASLLM communication assistance
Family communication	Phone calls and ad-hoc visits	Secure video calls and real-time messaging accessible through a web browser
Audit trail	Partial or manual logs; difficult to search and prepare for compliance reviews	Centralised audit logs that support search, export and compliance reporting
Setup	Complex IT deployment; staff	Web-based access with no installation

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	training may take significant time	required for staff or families
Data isolation	Shared databases; cross-facility data risk	Per-facility data management with role-based access control and secure database storage

The combination of SCVAM and ASLLM creates a unique value proposition by addressing both safety awareness and communication accessibility. These capabilities position Sphere Care as more than a documentation platform, transforming it into an intelligent care support system.

11. Design Methodology

Sphere Care was developed using a user-centred design approach. Throughout the project, the team focused on understanding the needs of aged care staff, family members, administrators, and compliance stakeholders before designing solutions. The design process followed three phases:

1. Research and Problem Definition

The research stage examined aged care documentation challenges, family communication needs, safety monitoring issues and compliance requirements in the Australian residential aged care context. Sources such as the Royal Commission Final Report, AIHW aged care statistics and existing aged care software products were reviewed to identify gaps in current systems.The key finding was that aged care staff are unlikely to adopt a system that increases their administrative workload. Therefore, Sphere Care was designed to reduce manual documentation, support faster incident review and improve communication without disrupting existing care routines.

2. Iterative Design and Prototyping

The staff web interface was designed with a simple, dashboard-based layout to support quick access to recordings, resident records, alerts and communication tools. A sidebar navigation pattern was used because it matches common clinical software conventions and reduces cognitive load for staff.
The client and family-facing interface was designed for users with varying levels of technical confidence. It uses large touch targets, clear labels, high-contrast text and simplified navigation to make key actions such as viewing updates, joining video calls and reading messages easier to access.

3. Implementation and Validation

Each sprint included a demonstration to the project supervisor. UI feedback was incorporated into the following sprint. The final UI was validated through end-to-end test scenarios based on the user personas above.

12. Design Trends and Approach

Sphere Care's visual design and UX draw on current industry trends in healthcare software and consumer applications:

- Clean, single-purpose screens: Each page is designed around one primary task, such as starting a recording, reviewing an alert or checking a resident update. This reduces cognitive load for staff working under time pressure in aged care settings.
- Human-in-the-loop AI design: AI-generated outputs, including flags, transcripts and summaries, are presented as suggestions that require staff review. The system is designed to support staff

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judgement rather than replace it, which is important for responsible AI use in clinical and care environments.
- Progressive disclosure: Advanced functions such as analytics, audit logs and retention settings are available when needed but are not shown prominently to frontline users. This keeps the interface simple for daily care tasks while still supporting administrative and compliance needs.
- Colour-coded severity levels: The flag severity system uses colour-coded categories to help staff quickly identify the urgency of an issue. This supports faster prioritisation of safety events while keeping alerts easy to understand.
- Offline-resilient communication: The system is designed to handle unstable network conditions by supporting reliable message delivery and reconnection logic. This is important for aged care facilities where Wi-Fi coverage may be inconsistent.
- Accessible by default: The staff interface uses responsive Bootstrap layouts, semantic HTML and ARIA attributes to improve accessibility. The mobile interface also follows accessibility principles such as clear typography, large touch targets and high-contrast display options.

13. Use Case Scenarios

The following use case scenarios are fictional but representative examples created to demonstrate how Sphere Care could support common aged care workflows. They are based on the project's target users, identified pain points and system features.

Scenario 1: The 2am Fall

A resident in a residential aged care facility wakes at 2am and attempts to get out of bed without assistance. Sphere Care detects a potential safety event and creates a high priority flag for staff review. The overnight nurse receives an alert, checks the relevant record and responds quickly . The event is stored with a timestamp, related resident information and review status, helping the facility maintain a clearer record of the incident.

Scenario 2: The Family Away From Home

David's mother lives in an aged care facility, but David lives interstate and cannot visit regularly. Through the Sphere Care mobile app, he can receive approved updates, view care summaries, message the care team and join video calls when needed. This helps him stay informed without requiring repeated phone calls to the facility.

Scenario 3: The Accreditation Audit

A facility is preparing for an accreditation audit. The administrator opens the Admin Console, selects the required data range and reviews relevant audit records, care sections, flags and system activity. Due to key actions are timestamped and linked to users and residents, the facility can prepare clearer evidence for compliance review.

14. Business Benefits

Sphere Care creates value for multiple stakeholder groups.

- **Benefits for Staff**
  - Reduced documentation workload.
  - Faster access to resident information.
  - Improved communication and collaboration.
  - Better visibility of care-related events.
- **Benefits for Residents**

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    - Safer care environments.
    - Improved communication accessibility.
    - Faster response to potential incidents.
    - Better continuity of care.
- **Benefits for Families**
    - Increased transparency.
    - Improved communication with care providers.
    - Greater confidence in care delivery.
- **Benefits for Facilities**
    - Improved operational efficiency.
    - Reduced administrative burden.
    - Enhanced compliance readiness.
    - Stronger documentation quality.

15.Compliance and Privacy

- **Consent tracking:** Resident and guardian consent status is tracked for each resident, with consent documents stored securely and linked to the relevant resident profile.
- **Data retention:** Retention periods are configurable for different data types, including media, transcripts and audit logs, in line with facility policy and aged care compliance requirements.
- **Soft deletion:** Deleted records are hidden from normal views but retained in the database for audit, recovery and compliance purposes.
- **Audit logging:** Key user and system actions are recorded in audit logs to support accountability and compliance review.
- **Role-based data access:** User permission and facility context are used to restrict access to relevant resident and care information.

16. Roadmap

Phase	Timeline	Features
MVP (v1.0)	Delivered	Recording, ASR transcription, AI-assisted flags, video/audio calls, messaging, audit logging, appointment bookings, records library, notifications and analytics
v2.0	6 months post-submission	Full EMR/FHIR integration, enhanced call recording with media tags, improved LLM retry logic, mobile push notifications and stronger transcript review tools
v3.0	12 months post-submission	AI- assisted predictive health alerts, multi-facility management console, enterprise SSO, advanced reporting, scalable AI worker deployment and enhanced consent management

17. Getting Started

- Install the required software: Python 3.10+, Node.js 18+, PostgreSQL and Git.
- Create and configure the PostgreSQL database named `sphere_care`.

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- Set up the backend virtual environment and install the required Python packages.
- Create the backend `.env` file with the database URL and LiveKit credentials.
- Start the FastAPI backend server using Uvicorn.
- Open the Staff Web Application at `http://localhost:8000`.
- Open the API documentation at `http://localhost:8000/docs`.
- Install the Mobile Client dependencies inside the `frontend_client` folder.
- Configure `frontend_client/.env` with the correct backend URL.
- Start the Mobile Client using Expo.
- Test the Mobile Client in a browser at `http://localhost:3000`, or scan the Expo QR code with Expo Go on an Android phone.
- For phone testing, make sure the phone and laptop are on the same WiFi network and use the laptop's WiFi IP address instead of `127.0.0.1`.

18. Conclusion

Australia's aged care sector faces a well-documented and growing crisis. Older Australians aged 65 and over are projected to make up between 21% and 23% of the total population by 2066, up from 15% in 2017 (ABS, 2018). At the same time, the sector faces a direct-care workforce shortfall projected to exceed 400,000 workers by 2050 if current trends continue (CEDA, 2021). The Royal Commission into Aged Care Quality and Safety has already documented the consequences of these pressures: inconsistent documentation, fragmented communication, and inadequate incident detection. Sphere Care was built to address these problems directly.

The platform brings together AI-assisted recording and transcription, real-time safety detection, secure family communication, and an immutable audit trail in a single web-based system. It is designed for the people who feel these pressures most: nursing staff who spend too much time on paperwork, family members who struggle to stay informed, and administrators who need audit-ready records on demand. Every design decision made during this project was grounded in those user needs, informed by the competitive landscape, and validated against Australian regulatory requirements including the Privacy Act 1988 and the Aged Care Quality Standards.

The MVP delivered in this project covers all core requirements. The four deferred items are documented with clear rationale and a prioritised roadmap. Sphere Care is ready for a limited pilot deployment and has a clear path to a full commercial product — one that gives back time to the people who deliver care, and peace of mind to the families who love them.

Sphere Care — Connecting care teams and families with the tools they deserve.

— End of Project Marketing —

University of Wollongong — CSIT321 Capstone Project — Sphere Care

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Individual Reflections

Name	Role	Contribution	Signature
Araf Akhter 8320561	Database and Data Model Developer	16.67%	ARAF
ChenCheng Lu 7316550	Backend API Developer	16.67%	ChenchengLu
Chingyu Chen 7910558	Client Mobile Application Developer	16.67%	ChingyuChen
Lite Ye 8326009	Backend Services Developer	16.67%	Lite Ye
Tonmoy Saker 8239083	AI and ML Developer	16.67%	
Yu Han Yong 8212041	Full Stack Developer	16.67%	

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GROUP Member 1 – Individual Reflection

Full Name	YuHan Yong
Student ID	8212041
Role in Project	Frontend Developer, Team Leader
Date Submitted	25/5/2026

Reflection on Learning

1 What did you learn technically?

Although I was already familiar with HTML and JavaScript before this project, Sphere Care pushed me to apply these skills in a much more complex, real-world context than I had previously experienced. The most significant technical growth for me was in API integration, learning how to connect a frontend interface to a FastAPI backend using fetch and managing different response states and error conditions cleanly across 13 different pages.

Beyond basic API calls. I gained deep experience with WebSocket connections for real-time functionality. Implementing live message delivery, real-time flag alerts, call and notifications required me to understand how WebSocket connections are established, maintained and recovered after disconnection. These concepts I had not worked on before this project. I also became much more confident working with Bootstrap at a production level, building a consistent, responsive layout system that could be shared across the entire staff web application.

2 What did you learn about working in a team?

Working across a six-person team with separate backend, mobile, AI and frontend responsibilities taught me a lot about the importance of agreed interfaces between components. The most challenging coordination was with the backend team. When API endpoints changed or new fields were added, I needed to update my fetch calls and UI rendering logic to match. This happened several times during development and highlighted the value of locking the API contract at the start of each sprint before implementation begins.

3 How has this project contributed to your professional development?

This project gave me my first experience building a complete, multi-page web application that connects to a real backend API with authentication, real-time communication, and role-based access control. These are exactly the kinds of systems that professional frontend developers work on in industry, and having delivered something of this scale gives me much greater confidence going into my career.

The experience of working under sprint deadlines, committing to a scope at the start of a sprint and delivering it by the review. It also gave me a real feel for Agile development in practice. I also discovered through this project that I enjoy frontend work and want to continue developing in this direction, particularly in building clean, user-focused interfaces for complex backend systems.

4 What would you do differently?

The area I would change most is the frontend-to-frontend-client call integration — specifically, the video and audio call flow between the Staff Web and the Client Mobile App through LiveKit. This was the most technically complex part of my work, requiring coordination between the call state machine on the backend, the WebSocket signalling layer, and the LiveKit token handling on both ends. If I were starting again, I would spend more time at the beginning of the project mapping out exactly how the call flow works end-to-end from the staff web initiating a call, through the backend issuing tokens, to the client mobile app accepting before writing any code. Having that clear picture upfront would have saved a significant amount of debugging time during integration.

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GROUP Member 2 – Individual Reflection

Full Name	Chencheng Lu
Student ID	7316550
Role in Project	Backend developer
Date Submitted	25/5/2026

Reflection on Learning

D.1 What did you learn technically?

I gained solid full-stack development experience focusing on video recording and media management systems.

On the backend side, I mastered standard RESTful API design with FastAPI. I learned to use async endpoints to handle time-consuming tasks such as starting and stopping video recording efficiently. I also got familiar with using dependency injection via Depends for unified authentication, separating session logic and file service reasonably. Besides, I learned to hide sensitive file access routes by setting "include_in_schema=False" to enhance interface security and protect private media resources.

On the frontend, I learned to optimize user experience with skeleton loading to avoid page layout shift during data loading. I mastered implementing drag-and-drop file upload functions with real-time preview, as well as designing multi-style view switching including grid, list and single modes. Meanwhile, I understood the benefits of modular JavaScript coding by splitting business logic into different JS files, which greatly improves code readability and later maintenance efficiency.

Overall, I learned how to build stable, secure and user-friendly production-level full-stack media service architectures.

2 What did you learn about working in a team?

From team cooperation, I have gained many valuable working insights. First of all, clear communication is essential. We need to timely share progress, discuss technical difficulties and align project goals, so that everyone can keep consistent work direction and avoid repeated work or misunderstanding.

Besides, I learned the importance of reasonable task division. Everyone can give full play to their strengths, such as being responsible for backend development, frontend design and function testing separately, which greatly improves overall work efficiency. I also realize the value of mutual support. When teammates meet technical obstacles, we discuss solutions together and share experience to solve problems faster. Moreover, listening to different ideas helps optimize project design and make products more perfect.

In addition, respecting team opinions and following unified work standards can keep the whole project orderly. In short, teamwork teaches me to cooperate actively, take responsibility and make joint efforts to finish tasks smoothly.

3 How has this project contributed to your professional development?

This project has greatly boosted my professional growth in multiple ways. Technically, I deepened my mastery of full-stack development. I became proficient in building efficient and secure APIs with FastAPI, mastering asynchronous programming, unified authentication and media file management logic. I also improved my frontend development skills, learning practical user experience optimization and standardized modular code writing, laying a solid foundation for developing production-level applications.

In terms of soft skills, cooperating with teammates taught me effective communication, reasonable task coordination and efficient problem-solving ability. I learned to sort out technical logic clearly and discuss solutions together to push progress forward steadily.

Besides, I formed stricter engineering thinking, paying more attention to code standardization, system security and actual user demands. All these practical experiences have made me more competent and

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calm in facing real development work, and greatly enhanced my comprehensive professional competitiveness.

4 What would you do differently?

First, I would plan the overall system architecture more carefully at the early stage. I will sort out all functional modules and data flow in advance to avoid later structure adjustments and repeated code revisions. Second, I would write more detailed unit tests for core backend interfaces and video recording functions. This can effectively reduce hidden bugs and make the system more stable in actual use. Third, I will optimize the front-end performance further. I plan to add file compression before uploading and optimize resource loading speed to bring smoother user experience. Besides, I will communicate more frequently with team members earlier to confirm requirements and unify development standards timely. In general, our team focuses on summarization and reflection after finishing tasks. Review shortcomings and optimize working methods constantly, so the whole team can cooperate more efficiently in future projects. I will pay more attention to early planning, code robustness and user experience details, making the whole project more complete, efficient and easier to maintain.

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GROUP Member 3 – Individual Reflection

Full Name	Lite Ye
Student ID	8326009
Role in Project	Backend Development
Date Submitted	25/5/2026

Reflection on Learning

1 What did you learn technically?

During this project, I improved my understanding of backend development and system integration. I learned how to design and manage backend services, connect APIs with databases, and handle data processing between different system modules. I also became more familiar with RESTful API development, asynchronous request handling, and backend logic implementation. Besides that, I gained practical experience in debugging, testing, and improving system stability and maintainability. Overall, this project strengthened my understanding of full-stack development workflows and gave me valuable experience in real-world backend development.

2 What did you learn about working in a team?

Through this project, I learned that communication and collaboration are very important in software development. Since team members worked on different parts of the system, we needed to discuss requirements, progress, and technical issues regularly to ensure smooth integration. I also learned the importance of task division and teamwork. By sharing responsibilities clearly and supporting each other when problems appeared, the team was able to complete the project more efficiently.In addition, discussing technical problems together helped improve the final system quality and strengthened my teamwork and communication skills.

3 How has this project contributed to your professional development?

This project improved both my technical and professional skills. Technically, I became more confident in backend development, API integration, database interaction, and system debugging. I also gained a better understanding of software engineering processes such as planning, implementation, testing, and system maintenance. Working in a team environment improved my communication, problem-solving, and collaboration abilities. Overall, this project gave me valuable practical experience and helped prepare me for future software development work.

4 What would you do differently?

If I could restart this project, I would spend more time planning the backend structure and system integration before development. Better planning could reduce repeated modifications and improve development efficiency.I would also focus more on testing and optimization during development to improve system stability and maintainability. In addition, I would communicate more frequently with teammates to ensure better consistency between frontend and backend development.Overall, I would place greater emphasis on early planning, testing, and system optimization to improve the overall project quality.

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GROUP Member 4 – Individual Reflection

Full Name	Chingyu Chen
Student ID	7910058
Role in Project	Mobile Developer/ Frontend Developer
Date Submitted	30/5/2026

Reflection on Learning

1 What did you learn technically?

Through this project, I learned how a real application is structured and developed beyond a simple classroom project. Before this project, I mostly understood programming as separate tasks, such as writing frontend pages or backend logic. However, this project helped me understand how different parts of a system work together. I learned how the mobile frontend communicates with the backend, how the backend connects to the database, and how user data can be stored, retrieved, and displayed in the app.

A major new area for me was mobile app development. Since most of my previous subjects focused more on general programming, web development, or theoretical concepts, building a mobile app gave me a very different experience. I learned more about screen design, navigation, API integration, device testing, and handling real user interactions on a phone.

I also gained knowledge about how AI models can be integrated into an application. This was challenging because AI is not only about the model itself, but also about how the model receives data from the app and returns useful results to users. Overall, this project helped me connect frontend, backend, database, and AI integration into one complete system.

2 What did you learn about working in a team?

I learned that teamwork requires both individual responsibility and clear communication. In our team, each member completed their own part, which helped the project move forward. Using GitHub also improved our teamwork because it allowed us to share code, track changes, and combine everyone's work more efficiently. However, I also realised that we could improve our discussion of feature details. Sometimes the general task was clear, but the exact user flow or function design was not clear enough, which caused extra changes later.

3 How has this project contributed to your professional development?

This project helped me develop skills that are useful for my future career. Since I mainly worked on the mobile app, I gained a deeper understanding of mobile app development, including interface design, API connection, testing, and debugging. I also learned that real software development is not only about coding, but also about planning, communication, and solving problems step by step. This experience made me more confident in working with different technologies and understanding how a real application is built.

4 What would you do differently?

If I started this project again with the same team and the same brief, I would spend more time on planning before writing code. At the beginning of the project, everyone should have a clear understanding of the project direction, target users, main functions, and expected user flow. This is important because if the team only has a general idea but not enough detail, the project direction can easily become unclear later.

Another thing I would change is the way we discuss feature details. Instead of only saying what each person is responsible for, we should confirm the exact requirements of each function, such as what the user can see, what actions they can perform, and how the system should respond. This would reduce misunderstandings and avoid reworking features later.

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GROUP Member 5 – Individual Reflection

Full Name	Araf Akhter
Student ID	8320561
Role in Project	Database and Data Model Developer
Date Submitted	31/5/2026

Reflection on Learning

1 What did you learn technically?

The most significant technical knowledge I gained from this project was database design and data modelling within a large-scale software system. Before this project, I had experience using relational databases in coursework, but Sphere Care allowed me to apply those concepts to a much more complex and realistic environment.

My primary responsibility was designing and maintaining the database structure used throughout the platform. This involved creating entity relationships, defining schemas, managing data consistency, and supporting communication between different modules. I learned how to convert business requirements into efficient database structures that are scalable, maintainable, and secure.

I also gained practical experience working with PostgreSQL, SQL queries, indexing strategies, relationship modelling, and database normalisation. Additionally, I learned how backend services interact with databases and how data flows between user interfaces, APIs, and storage systems. This project significantly improved my understanding of database architecture and demonstrated how well-designed data models contribute to the success of large software systems.

2 What did you learn about working in a team?

Working on Sphere Care taught me the importance of communication and coordination between different development teams. Because almost every system component depended on database structures, I needed to work closely with frontend, backend, and AI developers to ensure the database supported their requirements.

One important lesson I learned was that changes to database schemas can affect many other parts of a system. As a result, careful planning and regular communication were essential whenever modifications were required. Frequent discussions helped prevent inconsistencies and ensured that everyone worked with the same understanding of the data model.

I also learned the importance of flexibility. As requirements evolved throughout development, the database design needed to adapt while maintaining stability and performance. This experience strengthened my teamwork, communication, and collaboration skills while giving me valuable insight into how large development teams work together.

3 How has this project contributed to your professional development?

This project helped bridge the gap between academic learning and professional software development. I gained practical experience designing databases for a real-world application involving multiple user roles, communication systems, AI integrations, and reporting requirements.

Technically, I became more confident in database design, data modelling, query optimisation, and system integration. I also gained a better understanding of software engineering processes such as planning, implementation, testing, documentation, and maintenance.

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Beyond technical skills, the project improved my problem-solving abilities and taught me how to communicate technical decisions clearly with team members from different technical backgrounds. These experiences have strengthened my confidence and provided a strong foundation for future software engineering and database development roles.

4 What would you do differently?

If I were to start the project again, I would invest more time in planning the data model before implementation. While the database evolved successfully throughout development, creating more detailed relationship diagrams and scalability plans at the beginning would have reduced some later modifications.

I would also introduce more automated database testing and validation procedures to identify issues earlier and improve long-term maintainability. Additionally, I would document schema changes more thoroughly throughout the project to simplify integration and make the system easier for future developers to understand.

Overall, I would place greater emphasis on planning, testing, documentation, and scalability considerations to further improve the quality, reliability, and maintainability of the database architecture.

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GROUP Member 6 – Individual Reflection

Full Name	Tonmoy Sarker
Student ID	8239083
Role in Project	AI and ML Developer
Date Submitted	31.05.2026

Reflection on Learning

1 What did you learn technically?

The most significant technical knowledge I gained from this project was the design, development, and integration of artificial intelligence systems within a complete software platform. Before this project, I had experience training machine learning models and working with computer vision techniques, but I had not integrated AI into a large-scale multi-user application.

My primary responsibility was developing the AI components of Sphere Care, including SCVAM (Sphere Care Video Analysis Module) and ASLLM (AI Sign Language and Gesture Communication Module). Through SCVAM, I learned how to build video processing pipelines, perform event detection, manage AI-generated safety flags, and integrate computer vision models into real workflows. I gained practical experience using OpenCV, MediaPipe, PyTorch, and machine learning techniques for data preparation, model training, testing, and deployment.

For ASLLM, I learned how gesture recognition systems can be created using landmark extraction, feature engineering, and machine learning models. Building datasets, training models, improving recognition accuracy, and integrating AI outputs into the platform gave me valuable hands-on experience. Overall, this project significantly strengthened my understanding of applied AI, computer vision, machine learning deployment, and real-world AI integration.

2 What did you learn about working in a team?

This project taught me that successful AI development depends heavily on collaboration with other technical teams. Since the AI modules interacted with the backend, frontend, and mobile application, I had to communicate regularly with team members to ensure that AI outputs could be used effectively throughout the platform.

One challenge was ensuring that the outputs generated by SCVAM and ASLLM matched the data structures expected by the backend and user interfaces. To address this, we frequently discussed API requirements, testing results, and integration strategies. These discussions helped reduce compatibility issues and improved the overall quality of the system.

I also learned the importance of documentation and clear communication. Since team members worked on different components, maintaining shared understanding was essential for successful integration. This experience improved my teamwork, communication, and problem-solving skills while showing me how multidisciplinary teams collaborate to deliver a complete software product.

3 How has this project contributed to your professional development?

This project provided valuable experience that closely reflects modern software development practices. It allowed me to work on a real-world AI application involving machine learning, computer vision, backend integration, databases, and user interfaces.

Technically, I developed stronger skills in AI model development, dataset preparation, performance evaluation, and deployment. I also learned how to balance technical performance with practical

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business requirements and user needs. Professionally, I gained experience working within an Agile team environment, participating in sprint planning, progress reviews, and integration testing.

The project improved my confidence in presenting technical concepts, collaborating with developers from different disciplines, and solving complex technical problems. It reinforced my interest in artificial intelligence and has motivated me to continue pursuing AI-related development and research opportunities in the future.

4 What would you do differently?

If I were to restart this project, I would spend more time collecting and validating datasets during the early stages of development. Since AI performance depends heavily on data quality, larger and more diverse datasets would likely improve model accuracy and robustness.

I would also establish more structured testing procedures for the AI modules from the beginning of the project. Continuous evaluation throughout development would help identify issues earlier and reduce integration challenges later. Additionally, I would define clearer interfaces between the AI modules and backend services before implementation to simplify communication and reduce development time.

Overall, I would place greater emphasis on early planning, data quality, automated testing, and integration design to further improve both development efficiency and system performance.

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